College Access and Domestic Violence

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Domestic Violence Around the World

Violence Against Women: Deadliest place for women is at home, UN report says

UK police: Violence against women at 'epidemic' levels, threat is on par with terrorism

Alarm as US domestic violence killings remain high despite drop in homicides

The Alarming Rise of Gender-Based Violence in Brazil

What a breakfast murder in India says about attitudes to wife beating

Last month, police in India arrested a 46-year-old man who allegedly murdered his wife because his breakfast had too much salt.

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- Substantial social and economic costs:
  - Cost of violence against women: 2% of global GDP ≈ US$1.5 trillion (UN Women, 2016)
  - Women: reduced labor force participation, poor mental health and higher suicide risk (Bhattacharyya et al., 2011; Campbell, 2002)
  - Children: low birth weight, poor mental health, negative consequences on peers' learning and college outcomes (Aizer, 2011; Bhuller et al., 2024; Carrell et al., 2018)

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  - Newer Question: How does greater access to college affect domestic violence?

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- **Data:** National Family Health Survey (NFHS-4, 2015-2016), National Crime Records Bureau (police records)

Preview of Findings: Main Results

- Women are 5.2 p.p. less likely to experience domestic violence in the past 12 months (0.082 s.d.)
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- The effect is concentrated among younger women, who were more exposed to the program (8.9 p.p.)
- Fewer domestic violence cases reported to the police
  - Cruelty by Husband or His Relatives: ↓ 10.7 cases
  - Dowry Prohibition Act: ↓ 22.4 cases
  - Domestic Violence Act: ↓ 3.3 cases

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- Norms and empowerment: improvements in attitudes toward violence, ownership, household decision-making, freedom of women, husbands' gender attitudes
- Labor market: wife less likely to work; no effect on husband's employment
  - Household specialization: ↑ husband-only working couples, ↓ couples with both spouses working
  - Shift from low-skill to higher-skill jobs for both working wives and working husbands

Related Literature

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3. **Intra-household decision-making and labor supply**: Chiappori (1988); Chiappori et al. (2024); Duflo (2012); Chin (2012)
   - Contribution: new evidence on household specialization and occupational upgrading of husbands and wives

Road Map

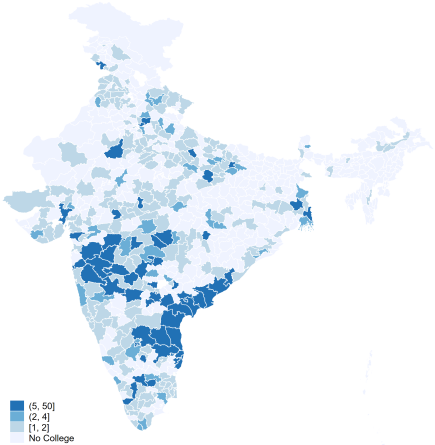
1. Introduction
2. Background
3. Data
4. Empirical Strategy
5. Results
6. Mechanisms
7. Conclusion

Background

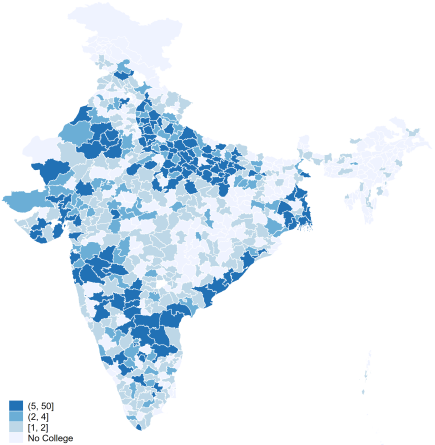
Background

- In 2008, the Government of India launched a program to provide grants to construct new colleges
- Eligibility for Grants:
  - Stage 1 (District level, 374): Districts with Gross Enrollment Ratio (GER) in higher education below the national average (12.4%, Census 2001)
  - Stage 2 (College level): located in an eligible district and set up on or after January 1, 2008
- Preferences are given to
  - Hilly/international border areas with higher minority/tribal populations
  - Areas lacking a college within 10 km
  - Rural areas with good transportation

Background



(a) No. of Colleges in 2001



(b) No. of Colleges in 2016

Data

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- College Data: 2021 All-India Survey on Higher Education (AISHE)
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  - Women Survey: household characteristics, education, health, employment, attitudes, domestic violence module
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- Administrative Data: National Crime Records Bureau (NCRB) at district level from 2013 to 2018

Measurement of Violence (NFHS-4)

(Does/did) your (last) husband ever do any of the following things to you? (Yes/No)

Physical Violence

- Push, shake, or throw objects at you
- Slap you
- Punch with fist or objects
- Kick, drag, or beat you
- Choke or burn you
- Threaten or attack with a weapon
- Forced unwanted sex
- Forced unwanted sexual acts

Emotional Violence

- Humiliate you in front of others
- Threaten harm to you or others
- Insult or degrade you

Any violence

***Three main outcomes**: any violence, physical violence, and emotional violence.*
Each outcome is measured as a dummy and an inverse-covariance weighted index (Anderson, 2008).

Empirical Strategy

Empirical Strategy

Regression Discontinuity Design: districts with 2001 GER below the national average (12.4%) were eligible for new colleges

$$Y_{id} = \alpha + \delta_t + \tau \cdot \text{Treated}_d + f(R_d - c) + \gamma X_{id} + \psi_s + \varepsilon_{id} \quad (1)$$

where

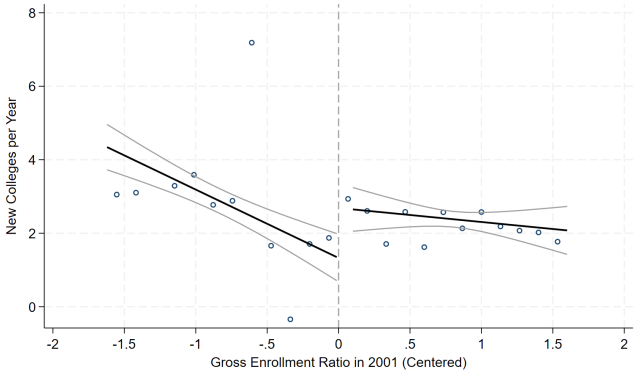
Y_{id} : outcome of interest for individual i in district d
R_d : district-level GER in 2001
c : national average GER (12.4%)
Treated_d : $= 1$ if district d is eligible ($R_d < c$)
δ_t, ψ_s : birth-year and state fixed effects
X_{id} : control variables
ε_{id} : error term

▸ McCrary density test ▸ Robustness of Manipulation Tests ▸ Covariate Discontinuity Test

“First-Stage”: Impact on Number of Colleges

- The program led to 1.2 additional colleges per year in treated districts

Figure Effects of the College Expansion on the Number of Colleges Constructed per Year (2009–2019)



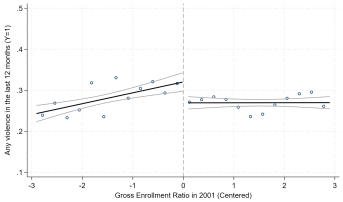
$\hat{\tau} = 1.210$ (robust p-value $= 0.001$) Mean $= 2.369$

▸ Placebo: pre-2008

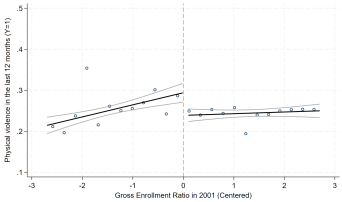
Results

Effects on Domestic Violence

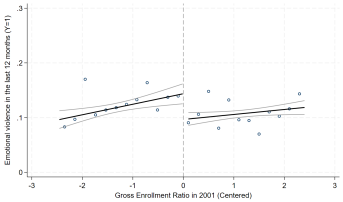
- The program ↓ the likelihood of experiencing any domestic violence in the last 12 months by 5.2 pp
- Physical violence ↓ by 4.7 pp and emotional violence ↓ by 3.5 pp Details



(a) Any Violence



(b) Physical Violence



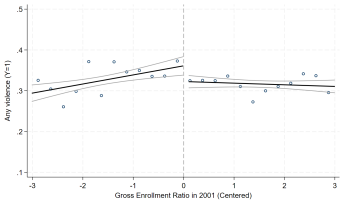
(c) Emotional Violence

Note: Domestic violence reported in the last 12 months against the Gross Enrollment Ratio. Optimal bandwidth and estimation following Calonico et al. (2014, 2022).

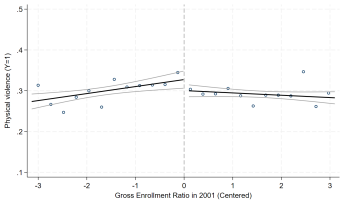
Effects on Domestic Violence

- Women in treated districts are 4.2 pp less likely to ever experience any domestic violence
  - Physical violence: ↓ 3.7 pp
  - Emotional violence: ↓ 2.5 pp

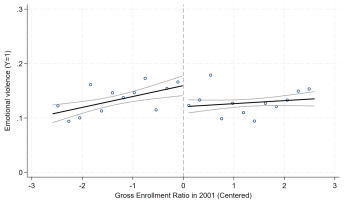
Details



(a) Any Violence



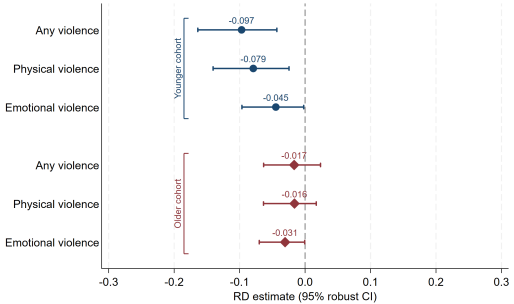
(b) Physical Violence



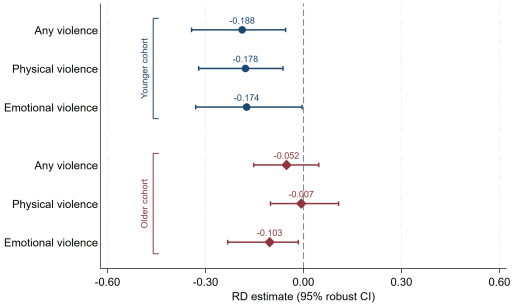
(c) Emotional Violence

Note: Ever-reported domestic violence against the Gross Enrollment Ratio. Optimal bandwidth and estimation following Calonico et al. (2014, 2022).

Heterogeneity Analysis



(a) Dummies



(b) Indices

Note: The younger cohort includes women under 30 years old while the older cohort consists of women at least 30 years old. Points show conventional RD estimates; lines show 95% robust bias-corrected confidence intervals (Calonico et al., 2014, 2022).

Evidence from Filed Police Reports

	Insult to Women's Modesty	Cruelty by Husband or His Relatives	Dowry Prohibition Act	Domestic Violence Act
	(1)	(2)	(3)	(4)
RD Estimate	3.974	-10.709	-22.447	-3.268
	(4.946)	(18.677)	(4.320)	(1.042)
	[0.225]	[0.029]	[0.000]	[0.000]
Mean Dep Var	14.759	164.849	15.548	0.743
Bandwidth	2.758	1.943	1.557	3.797
Effective observations	1268	923	607	1500

Note: Data are from National Crime Records Bureau (NCRB) from 2013 to 2018. *Insult to Women's Modesty* refers to registered cases involving non-physical harassment of women. *Cruelty by husband or his relatives* refers to registered cases of domestic violence, including emotional, mental, and physical abuse. *Dowry Prohibition Act* refers to registered cases under the Dowry Prohibition Act, 1961, involving the giving, taking, demanding, or advertising of dowry. *Domestic Violence Act* refers to registered cases under the Protection of Women from Domestic Violence Act, 2005, involving applications for civil remedies, including protection orders, residence orders, monetary relief, custody orders, and compensation orders. All specifications include year and state fixed effects.

Mechanisms

Education

Table Effects of the College Expansion on Years of Education

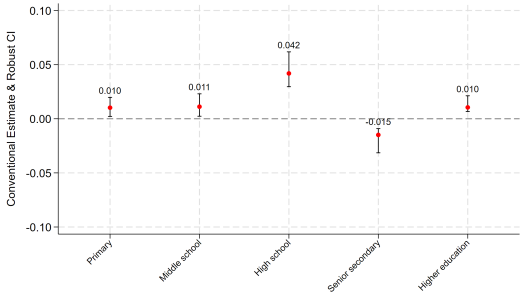
	Years of education			
	All women (1)	**All men** (2)	**Wives** (3)	**Husbands** (4)
RD Estimate	0.565	0.190	0.440	0.563
	(0.097)	(0.141)	(0.195)	(0.183)
	[0.000]	[0.130]	[0.039]	[0.000]
Mean Dep Var	6.838	8.373	6.103	7.666
Bandwidth	1.514	2.813	2.417	2.669
Effective observations	97233	35421	21044	22981

Note: Data are from NFHS-4. Column (1) includes all women aged 18–46; column (2) all men; columns (3) and (4) married women and their husbands. All specifications control for religion and include birth-year and state fixed effects.

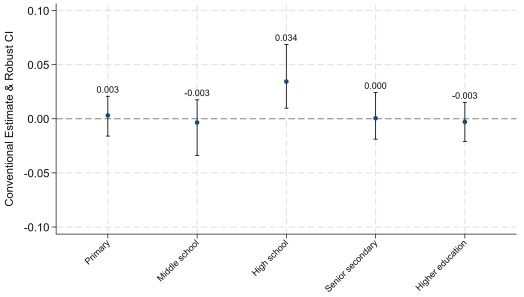
Education: All Sample

- Women: ↑ primary, middle, and high school completion, and higher education enrollment
- Men: ↑ high school completion

Details



(a) All Women

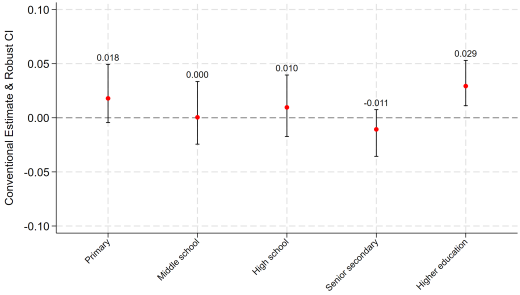


(b) All Men

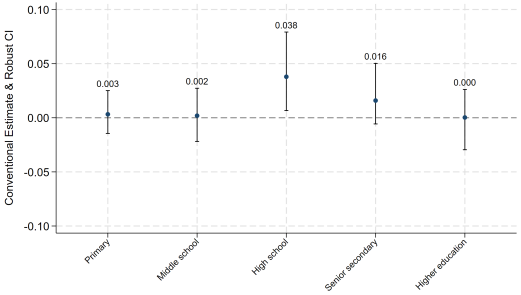
Education: Couples

- Wives: 2.9 pp ↑ in higher education enrollment
- Husbands: 3.8 pp ↑ in high school completion

Details



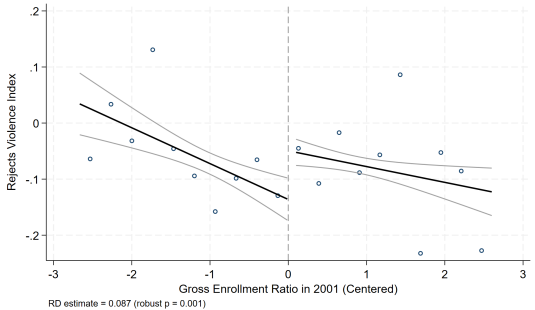
(a) Wives

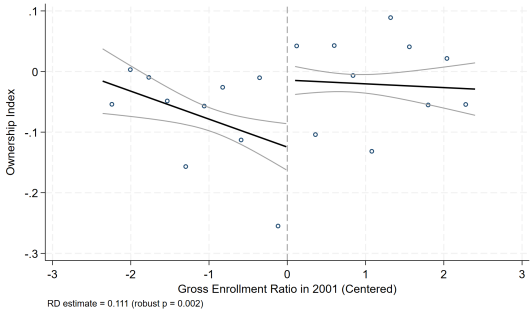


(b) Husbands

Women Empowerment and Autonomy

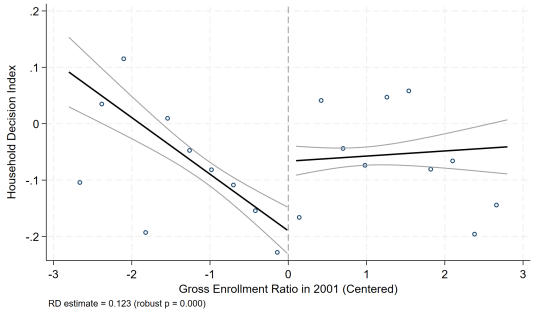
- Women's rejection of domestic violence ↑ by 0.087 SD in treated districts ▸ By item
- Women's ownership of assets (house, land, mobile phone) and bank accounts ↑ 0.111 SD in treated districts ▸ By item

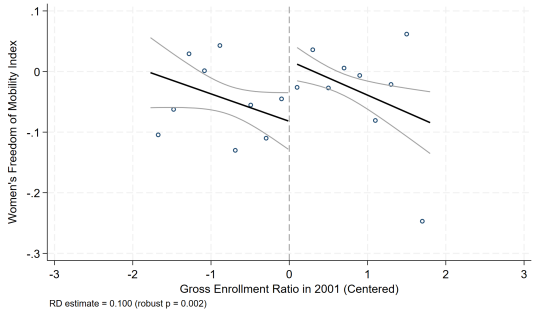




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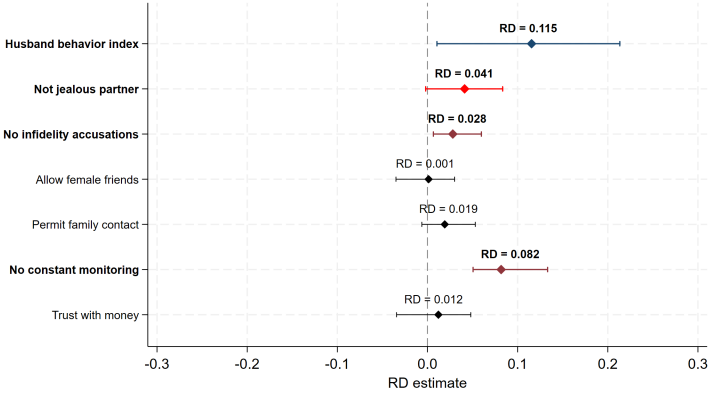
- Women's household decision-making ↑ by 0.123 SD in treated districts (By item)
- Women's freedom of movement ↑ 0.1 SD in treated districts (By item)





Husbands' Behavior

- Husband behavior index ↑ 0.115 SD in treated districts
- Husbands are more likely to trust their partners, grant them greater freedom, have less controlling behaviors



Employment and Occupational Choices

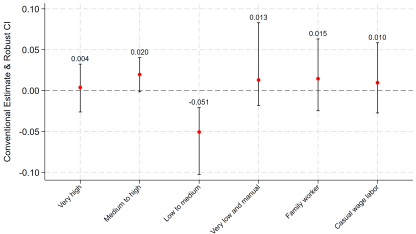
Table Effects of the College Expansion on Couples' Employment

	Husband Working	Wife Working	Only Husband Working	Only Wife Working	Both Working
	(1)	(2)	(3)	(4)	(5)
RD Estimate	0.008	-0.050	0.050	-0.002	-0.049
	(0.007)	(0.018)	(0.019)	(0.003)	(0.017)
	[0.457]	[0.004]	[0.007]	[0.638]	[0.002]
Mean Dep Var	0.957	0.299	0.650	0.009	0.310
Bandwidth	3.068	2.402	2.378	2.686	2.411
Effective observations	29761	22964	22851	25206	22964

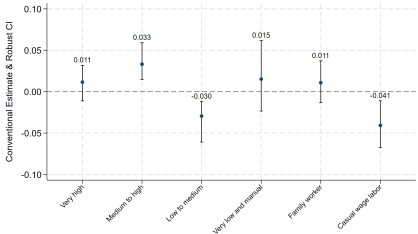
Note: Data are from NFHS-4. *Husband Working* and *Wife Working* are indicators equal to 1 if the husband or wife is currently employed, respectively, and 0 otherwise. *Only Husband*, *Only Wife*, and *Both Working* indicate households in which only the husband is employed, only the wife is employed, or both spouses are employed, respectively.

Employment and Occupational Choices

- Working wives: ↑ 2.0 pp in medium- to high-skilled occupations; ↓ 5.1 pp in low- to medium-skilled occupations
- Working husbands: ↑ 3.3 pp in medium- to high-skilled occupations; ↓ 3.0 pp in low- to medium-skilled occupations and ↓ 4.1 pp in casual agricultural wage labor



(a) Working wives



(b) Working husbands

Notes: Data is from NFHS-4. *Very High Skill*, *Medium to High Skill*, *Low to Medium Skill*, and *Very Low and Manual Skill* are indicator variables equal to 1 if an individual works in an occupation classified within the corresponding skill category, and 0 otherwise. *Very High Skill* includes occupations such as scientists, economists, and jurists. *Medium to High Skill* includes occupations such as transport and communication supervisors, technical salesmen, and commercial travelers. *Low to Medium Skill* includes occupations such as sales workers, service workers, and hotel and restaurant keepers. *Very Low and Manual Skill* includes occupations such as plumbers, drivers, and blacksmiths. *Family Worker (Agriculture)* is an indicator variable equal to 1 if an individual works as a cultivator or farmer, and 0 otherwise. *Casual Wage Labor* is an indicator variable equal to 1 if an individual works as an agricultural, forestry, or hunting laborer, and 0 otherwise.

Other Channels

	Marriage Market		Fertility Preferences		
	Age Gap (H–W) (1)	Educ. Gap (H–W) (2)	Ideal No. Children (3)	Contraception Use Pref. (4)	Husband's Drinking habits (5)
RD Estimate	0.169	0.100	0.012	0.011	-0.033
	(0.172)	(0.158)	(0.026)	(0.006)	(0.019)
	[0.331]	[0.248]	[0.664]	[0.021]	[0.102]
Mean Dep Var	4.878	1.563	0.206	0.915	0.316
Bandwidth	3.084	2.648	2.618	2.591	3.013
Effective observations	26166	22679	19152	65485	19693

Note: Data is from NFHS-4. *Age Gap (H–W)* is the age difference between the husband and the wife. *Educ. Gap (H–W)* is the difference in years of schooling between the husband and the wife. *Ideal No. of Children* is the difference between the ideal number of sons and daughters reported by the husband. *Contraception Use Pref.* is an indicator equal to 1 if the decision to use contraception is made mainly by the wife or jointly by the couple, and 0 otherwise. *Husband Drinks* is an indicator equal to 1 if the husband consumes alcohol, and 0 otherwise.

Robustness Checks

- **Specification:** alternative kernels, bandwidths, and polynomial orders
  - NFHS-4
    - full sample ▸ Full sample
    - by age cohort ▸ Younger cohort ▸ Older cohort
  - NCRB administrative data ▸ Robustness 1 ▸ Robustness 2
- **Marriage migration:** restrict NFHS-4 to women living in the same residence
  - at least 10 years ▸ +10 years
  - at least 20 years ▸ +20 years
- **Level of treatment:** collapse NFHS-4 to the district level ▸ Younger cohort ▸ Older cohort

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  - ~~Higher women's employment~~
- **Key takeaway**: Investments in higher education infrastructure and access can generate benefits that extend well beyond labor-market outcomes

Thank you!

Ha Luong

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Appendix Empirical Strategy

Figure McCrary density test (robust p-value = 0.419)



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Appendix Empirical Strategy: Robustness of Manipulation Tests using Different Kernels and Bandwidth

	T-statistic	P-value	Effective N
Panel A: Kernel Choice			
Triangular (baseline)	-0.808	0.419	286
Uniform	-0.733	0.463	273
Epanechnikov	-0.717	0.474	283
Panel B: Bandwidth Selection			
Difference MSE (hl≠hr)	-0.784	0.433	279
Sum MSE	-0.821	0.412	303
MSE by side	-0.405	0.685	340
Panel C: Polynomial Order			
Higher-order (p=3, q=4)	-0.550	0.582	389

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Appendix Empirical Strategy: Covariate Discontinuity Test

Variable	MSE-Optimal Bandwidth	RD Estimates	Robust Inference p-value	Robust Inference Conf. Int.	Effective No. of Obs.
Population (log)	3.153	-0.080	0.841	[-0.839, 0.683]	207
Scheduled Caste Population	3.117	-0.426	0.339	[-1.48, 0.51]	203
Scheduled Tribe Population	3.878	-0.967	0.282	[-3.21, 0.93]	221
Total Expenditure	3.605	-0.458	0.409	[-1.54, 0.63]	242
Number of Colleges	3.703	-3.18	0.440	[-13.1, 5.7]	248
Pucca Road Length	2.738	226	0.249	[-194, 750]	172
Road Lighting Electrical Conn.	3.789	-5879	0.570	[-28566, 15725]	250
Households	3.279	-0.185	0.588	[-0.980, 0.555]	215
Area (km^2)	2.942	22.3	0.572	[-64.5, 116.7]	194
Literacy Rate	3.006	-0.004	0.859	[-0.085, 0.071]	196
Female Share	3.725	0.002	0.735	[-0.009, 0.013]	248
Female Literacy Rate	2.878	-0.007	0.842	[-0.105, 0.085]	186
Male Literacy Rate	3.423	0.000	0.929	[-0.065, 0.060]	225
Electricity (all purposes)	3.746	-0.011	0.729	[-0.124, 0.087]	248
No. of Primary Schools	3.122	61.8	0.872	[-507, 598]	204
No. of Middle Schools	3.080	-25.8	0.560	[-219, 119]	199
No. of Secondary Schools	3.454	-22.6	0.496	[-110, 54]	228
No. of Senior Secondary Schools	3.702	-4.76	0.815	[-21.8, 17.1]	248
No. of Hospitals	2.705	9.61	0.428	[-10.5, 24.8]	172
Tap Water	3.535	-184	0.570	[-750, 413]	236
Average Night Lights (1994-2007)	3.176	-2025	0.889	[-16014, 13888]	226
Total Night Lights (1994-2007)	3.176	-52649	0.889	[-416377, 361084]	226
Average Forest Cover (2001-2007)	3.094	-3.56	0.254	[-12.7, 3.3]	217

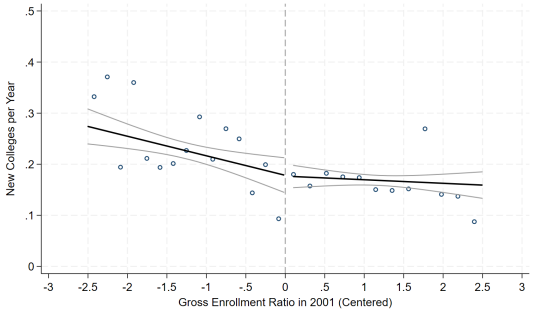
Notes: Demographic data is from the 2001 Population Census. Nighttime light intensity for 1994–2007 comes from the DMSP-OLS annual composite imagery. Forest cover data (2001–2007) is derived from the MODIS Vegetation Continuous Fields (VCF) product (Asher et al., 2021).

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Appendix Empirical Strategy

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Figure Number of colleges Constructed per Year (pre-2008)



$\hat{\tau} = -0.021$ (robust p-value $= 0.458$) Mean $= 0.173$

Appendix Main Results

Table Effects of the College Expansion on Domestic Violence (last 12 months)

	Any Violence		Physical Violence		Emotional Violence	
	Dummy (1)	Index (2)	Dummy (3)	Index (4)	Dummy (5)	Index (6)
RD Estimate	-0.052	-0.097	-0.047	-0.079	-0.035	-0.128
	(0.019)	(0.038)	(0.019)	(0.039)	(0.015)	(0.052)
	[0.003]	[0.005]	[0.007]	[0.041]	[0.019]	[0.012]
Mean Dep Var	0.272	0.006	0.246	0.006	0.108	0.009
Bandwidth	2.931	4.053	2.730	4.044	2.469	2.414
Effective observations	18215	25950	16276	24050	15241	15000

Note: Data is from NFHS-4. All specifications control for religion and include birth-year and state fixed effects.

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Appendix Main Results

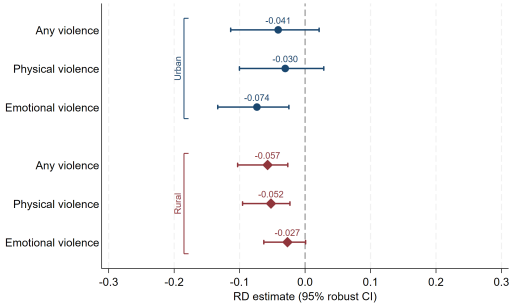
Table Effects of the College Expansion on Domestic Violence (Reported Ever)

	Any Violence		Physical Violence		Emotional Violence	
	Dummy (1)	Index (2)	Dummy (3)	Index (4)	Dummy (5)	Index (6)
RD Estimate	-0.042	-0.082	-0.037	-0.074	-0.025	-0.091
	(0.019)	(0.040)	(0.018)	(0.039)	(0.015)	(0.045)
	[0.012]	[0.027]	[0.017]	[0.049]	[0.051]	[0.019]
Mean Dep Var	0.317	0.005	0.295	0.004	0.126	0.009
Bandwidth	3.043	3.459	3.142	3.602	2.614	2.850
Effective observations	19693	23088	20718	24779	16520	18697

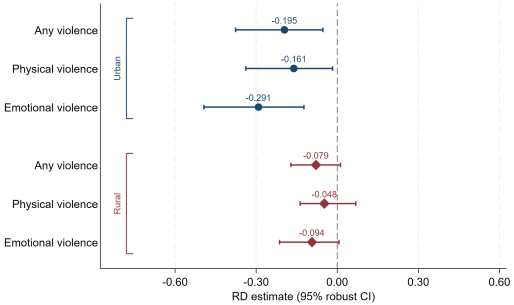
Note: Data is from NFHS-4. All specifications control for religion and include birth-year and state fixed effects.

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Appendix Main Results



(a) Dummies



(b) Indices

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Appendix Main Results

Table Robustness of Effects of the College Expansion on Any Domestic Violence Using Different Kernel, Bandwidths, and Polynomial Orders

	(1)	(2)	(3)	(4)	(5)
Panel A: Any Violence (Dummy)					
RD Estimate	-0.045	-0.049	-0.051	-0.042	-0.044
	(0.023)	(0.020)	(0.019)	(0.022)	(0.024)
	[0.062]	[0.010]	[0.004]	[0.054]	[0.130]
Mean Dep Var	0.272	0.272	0.272	0.272	0.272
Bandwidth	1.637	2.441	2.721	1.577	3.595
Effective observations	9641	14517	16315	9447	22417
Panel B: Any Violence (Index)					
RD Estimate	-0.096	-0.110	-0.099	-0.081	-0.096
	(0.051)	(0.041)	(0.038)	(0.046)	(0.060)
	[0.026]	[0.003]	[0.005]	[0.071]	[0.138]
Mean Dep Var	0.006	0.006	0.006	0.006	0.006
Bandwidth	1.869	3.220	3.977	2.298	3.525
Effective observations	11284	21094	25358	14423	23418
Bandwidth selection	$\hat{h}_{MSE}$	$\hat{h}_{MSE}$	$\hat{h}_{MSESUM}$	$\hat{h}_{CERSUM}$	$\hat{h}_{MSE}$
Local Polynomial Degree	1	1	1	1	2
Kernel	Uniform	Epanechnikov	Triangular	Triangular	Triangular

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Appendix Main Results

Table Robustness of Effects of the College Expansion on Any Domestic Violence Using Different Kernel, Bandwidths, and Polynomial Orders (Younger cohort)

	(1)	(2)	(3)	(4)	(5)
Panel A: Younger Cohort (Dummy)					
RD Estimate	-0.093	-0.098	-0.097	-0.078	-0.087
	(0.035)	(0.031)	(0.030)	(0.035)	(0.039)
	[0.010]	[0.001]	[0.001]	[0.021]	[0.054]
Mean Dep Var	0.271	0.271	0.271	0.271	0.271
Bandwidth	1.879	2.591	2.807	1.701	3.524
Effective observations	4422	6227	7075	4136	9085
Panel B: Younger Cohort (Index)					
RD Estimate	-0.138	-0.187	-0.192	-0.154	-0.184
	(0.082)	(0.074)	(0.071)	(0.083)	(0.095)
	[0.054]	[0.007]	[0.004]	[0.057]	[0.060]
Mean Dep Var	0.004	0.004	0.004	0.004	0.004
Bandwidth	1.704	2.613	3.014	1.823	3.506
Effective observations	4291	6500	7750	4583	9389
Bandwidth selection	$\hat{h}_{MSE}$	$\hat{h}_{MSE}$	$\hat{h}_{MSESUM}$	$\hat{h}_{CERSUM}$	$\hat{h}_{MSE}$
Local Polynomial Degree	1	1	1	1	2
Kernel	Uniform	Epanechnikov	Triangular	Triangular	Triangular

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Appendix Main Results

Table Robustness of Effects of the College Expansion on Any Domestic Violence Using Different Kernel, Bandwidths, and Polynomial Orders (Older cohort)

	(1)	(2)	(3)	(4)	(5)
Panel C: Older Cohort (Dummy)					
RD Estimate	-0.026	-0.018	-0.017	-0.017	-0.013
	(0.029)	(0.024)	(0.023)	(0.027)	(0.029)
	[0.354]	[0.368]	[0.416]	[0.488]	[0.824]
Mean Dep Var	0.272	0.272	0.272	0.272	0.272
Bandwidth	1.707	2.931	3.140	1.867	4.132
Effective observations	5982	10821	11528	6318	15122
Panel D: Older Cohort (Index)					
RD Estimate	-0.081	-0.054	-0.050	-0.024	-0.049
	(0.056)	(0.050)	(0.053)	(0.063)	(0.071)
	[0.173]	[0.266]	[0.335]	[0.696]	[0.538]
Mean Dep Var	0.007	0.007	0.007	0.007	0.007
Bandwidth	2.541	3.662	3.384	2.007	4.078
Effective observations	9579	14569	13407	7522	15601
Bandwidth selection	$\hat{h}_{MSE}$	$\hat{h}_{MSE}$	$\hat{h}_{MSESUM}$	$\hat{h}_{CERSUM}$	$\hat{h}_{MSE}$
Local Polynomial Degree	1	1	1	1	2
Kernel	Uniform	Epanechnikov	Triangular	Triangular	Triangular

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Appendix Main Results

Table Robustness: NCRB Crimes Against Women (1/2)

	Crimes Against Women (NCRB)				
	(1)	(2)	(3)	(4)	(5)
Panel A: Insult to Women's Modesty					
RD Estimate	-9.490	11.393	10.175	-3.919	-5.119
	(5.554)	(4.885)	(4.760)	(4.834)	(7.382)
	[0.188]	[0.009]	[0.013]	[0.501]	[0.255]
Mean Dep Var	14.759	14.759	14.759	14.759	14.759
Bandwidth	1.503	3.224	3.410	2.251	2.202
Effective observations	722	1535	1619	1073	1063
Panel B: Cruelty by Husband or His Relatives					
RD Estimate	-48.952	8.404	-7.872	44.419	8.244
	(21.055)	(19.552)	(19.169)	(18.426)	(34.258)
	[0.000]	[0.519]	[0.030]	[0.134]	[0.694]
Mean Dep Var	164.849	164.849	164.849	164.849	164.849
Bandwidth	1.295	2.088	1.845	1.218	2.796
Effective observations	609	979	857	609	1280
Bandwidth selection	$\hat{h}_{MSE}$	$\hat{h}_{MSE}$	$\hat{h}_{MSESUM}$	$\hat{h}_{CERSUM}$	$\hat{h}_{MSE}$
Local Polynomial Degree	1	1	1	1	2
Kernel	Uniform	Epanechnikov	Triangular	Triangular	Triangular

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Appendix Main Results

Table Robustness: NCRB Crimes Against Women (2/2)

	Crimes Against Women (NCRB)				
	(1)	(2)	(3)	(4)	(5)
Panel C: Dowry Prohibition Act					
RD Estimate	-18.392	-22.832	-15.917	-22.799	-26.449
	(5.059)	(4.724)	(4.426)	(4.113)	(4.788)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Mean Dep Var	15.548	15.548	15.548	15.548	15.548
Bandwidth	1.661	1.542	2.102	1.400	2.635
Effective observations	647	607	853	563	1007
Panel D: Domestic Violence Act					
RD Estimate	-3.824	-3.655	-3.548	-1.327	-2.734
	(1.127)	(1.129)	(0.834)	(0.406)	(0.516)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Mean Dep Var	0.743	0.743	0.743	0.743	0.743
Bandwidth	2.431	3.150	2.470	1.645	3.795
Effective observations	942	1253	947	647	1500
Bandwidth selection	$\hat{h}_{MSE}$	$\hat{h}_{MSE}$	$\hat{h}_{MSESUM}$	$\hat{h}_{CERSUM}$	$\hat{h}_{MSE}$
Local Polynomial Degree	1	1	1	1	2
Kernel	Uniform	Epanechnikov	Triangular	Triangular	Triangular

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Appendix Main Results

Table Effects of the College Expansion on Domestic Violence, Women in Same Residence for 10+ Years

	Any Violence		Physical Violence		Emotional Violence	
	Dummy (1)	Index (2)	Dummy (3)	Index (4)	Dummy (5)	Index (6)
Panel A: Reported Ever						
RD Estimate	-0.055	-0.077	-0.053	-0.067	-0.029	-0.100
	(0.027)	(0.049)	(0.026)	(0.051)	(0.021)	(0.063)
	[0.017]	[0.086]	[0.015]	[0.141]	[0.108]	[0.057]
Mean Dep Var	0.344	0.039	0.321	0.041	0.134	0.030
Bandwidth	2.728	3.898	2.832	3.585	2.616	2.861
Effective observations	9624	14039	10298	13033	9096	10298
Panel B: Reported in the last 12 months						
RD Estimate	-0.055	-0.105	-0.053	-0.066	-0.040	-0.149
	(0.027)	(0.054)	(0.026)	(0.061)	(0.021)	(0.072)
	[0.023]	[0.024]	[0.022]	[0.275]	[0.035]	[0.031]
Mean Dep Var	0.290	0.023	0.263	0.030	0.113	0.022
Bandwidth	2.687	3.677	2.645	3.066	2.543	2.513
Effective observations	8591	13254	8409	9642	8755	8698

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Appendix Main Results

Table Effects of the College Expansion on Domestic Violence, Women in Same Residence for 20+ Years

	Any Violence		Physical Violence		Emotional Violence	
	Dummy (1)	Index (2)	Dummy (3)	Index (4)	Dummy (5)	Index (6)
Panel A: Reported Ever						
RD Estimate	-0.105	-0.126	-0.130	-0.117	-0.056	-0.134
	(0.044)	(0.085)	(0.046)	(0.085)	(0.033)	(0.087)
	[0.003]	[0.065]	[0.001]	[0.088]	[0.030]	[0.043]
Mean Dep Var	0.339	0.044	0.316	0.042	0.138	0.046
Bandwidth	2.760	4.075	2.376	4.093	2.882	3.812
Effective observations	3910	5807	3416	5825	4229	5578
Panel B: Reported in the last 12 months						
RD Estimate	-0.134	-0.237	-0.165	-0.164	-0.068	-0.214
	(0.047)	(0.109)	(0.049)	(0.105)	(0.034)	(0.114)
	[0.001]	[0.015]	[0.000]	[0.049]	[0.030]	[0.038]
Mean Dep Var	0.284	0.019	0.257	0.022	0.115	0.028
Bandwidth	2.254	2.986	1.975	3.467	2.426	2.606
Effective observations	2999	4287	2538	4579	3332	3549

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Appendix Main Results

Table Effects of the College Expansion on Domestic Violence at District Level, Younger Cohort

	Reported Ever			Reported in the last 12 months		
	Any Violence (1)	**Physical Violence** (2)	**Emotional Violence** (3)	**Any Violence** (4)	**Physical Violence** (5)	**Emotional Violence** (6)
RD Estimate	-5.473	-4.661	-2.169	-5.038	-4.220	-2.506
	(1.957)	(1.883)	(0.760)	(1.596)	(1.543)	(0.678)
	[0.003]	[0.010]	[0.002]	[0.001]	[0.004]	[0.000]
Mean Dep Var	11.213	10.381	4.386	9.580	8.683	3.835
Bandwidth	2.578	2.568	2.227	2.627	2.563	2.235
Effective observations	177	177	158	180	177	159

Notes: Data are from NFHS-4. *Any violence* is the district-level number of women who have ever experienced either physical or emotional violence by their husband or partner. *Physical violence* is the district-level number of women who have ever been pushed, shaken, slapped, punched, kicked, dragged, strangled, burned, threatened with a weapon, or had their arm twisted or hair pulled by their husband or partner. *Emotional violence* is the district-level number of women who have ever been humiliated, threatened with harm, insulted, or made to feel bad by their husband or partner.

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Appendix Main Results

Table Effects of the College Expansion on Domestic Violence at District Level, Older Cohort

	Reported Ever			Reported in the last 12 months		
	Any Violence (1)	**Physical Violence** (2)	**Emotional Violence** (3)	**Any Violence** (4)	**Physical Violence** (5)	**Emotional Violence** (6)
RD Estimate	-1.106	-1.264	-1.626	-1.987	-1.807	-1.986
	(1.937)	(1.951)	(0.941)	(1.464)	(1.487)	(0.998)
	[0.702]	[0.643]	[0.039]	[0.082]	[0.153]	[0.032]
Mean Dep Var	18.169	16.902	7.290	14.024	12.630	5.943
Bandwidth	2.934	2.914	3.492	3.476	3.160	2.932
Effective observations	213	213	252	251	228	213

Notes: Data are from NFHS-4. *Any violence* is the district-level number of women who have ever experienced either physical or emotional violence by their husband or partner. *Physical violence* is the district-level number of women who have ever been pushed, shaken, slapped, punched, kicked, dragged, strangled, burned, threatened with a weapon, or had their arm twisted or hair pulled by their husband or partner. *Emotional violence* is the district-level number of women who have ever been humiliated, threatened with harm, insulted, or made to feel bad by their husband or partner.

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Appendix Mechanisms

Table Effects of the College Expansion on Educational Attainment

	Years of education	Primary school	Middle school	High school	Senior sec. school	Higher edu.
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
RD Estimate	0.565	0.010	0.011	0.042	-0.015	0.010
	(0.097)	(0.004)	(0.005)	(0.008)	(0.006)	(0.004)
	[0.000]	[0.015]	[0.016]	[0.000]	[0.000]	[0.000]
Mean Dep Var	6.838	0.127	0.156	0.181	0.128	0.127
Bandwidth	1.514	2.946	2.609	1.292	2.001	3.854
Effective observations	97233	193854	163763	83427	128635	251900
Panel B: Men						
RD Estimate	0.190	0.003	-0.003	0.034	0.000	-0.003
	(0.141)	(0.009)	(0.013)	(0.015)	(0.011)	(0.009)
	[0.130]	[0.800]	[0.532]	[0.009]	[0.807]	[0.750]
Mean Dep Var	8.373	0.126	0.179	0.244	0.159	0.156
Bandwidth	2.813	3.070	2.326	2.277	3.253	4.212
Effective observations	35421	37419	28686	27471	40706	52666

Note: Data is from NFHS-4. The sample consists of women between the ages of 18 and 46. We include religion as a control and birth year and state fixed effects.

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Appendix Mechanisms

Table Effects of the College Expansion on Couple's Educational Attainment

	Years of edu. (1)	Primary school (2)	Middle school (3)	High school (4)	Senior sec. school (5)	Higher edu. (6)
Panel A: Wife's Education						
RD Estimate	0.440	0.018	0.000	0.010	-0.011	0.029
	(0.195)	(0.014)	(0.015)	(0.014)	(0.011)	(0.011)
	[0.039]	[0.102]	[0.758]	[0.445]	[0.202]	[0.003]
Mean Dep Var	6.103	0.142	0.169	0.178	0.094	0.094
Bandwidth	2.417	2.602	2.677	2.852	3.124	2.937
Effective observations	21044	22574	22981	25546	28101	26517
Panel B: Husbands/Partner's Education						
RD Estimate	0.563	0.003	0.002	0.038	0.016	0.000
	(0.183)	(0.010)	(0.013)	(0.019)	(0.014)	(0.014)
	[0.000]	[0.598]	[0.830]	[0.020]	[0.119]	[0.907]
Mean Dep Var	7.666	0.144	0.177	0.236	0.123	0.132
Bandwidth	2.669	4.086	3.426	2.169	2.398	2.436
Effective observations	22981	36362	31264	19290	20940	21143

Note: Data is from NFHS-4. The sample consists of women between the ages of 18 and 46. We include religion as a control and birth year and state fixed effects.

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Appendix Mechanisms

Table Effects of the College Expansion on Women's Attitudes Toward Violence

	Goes Out Without Telling	Neglects Children	Argues with Husband	Refuses Sex	Poor Cooking
RD Estimate	0.029	0.025	0.015	0.019	0.031
	(0.014)	(0.015)	(0.017)	(0.011)	(0.013)
	[0.019]	[0.024]	[0.520]	[0.063]	[0.004]
Mean Dep Var	0.760	0.703	0.731	0.869	0.821
Bandwidth	2.399	2.564	1.848	2.830	2.352
Effective observations	26424	28008	19641	31834	26271

Notes: Data is from NFHS-4. Each column reports results from a separate regression. *Goes Out without Telling*, *Neglects Children*, *Argues with Husband*, *Refuses Sex*, and *Poor Cooking* are indicators equal to 1 if the respondent reports that a husband is not justified in beating his wife, respectively, for going out without telling him, neglecting the children, arguing with him, refusing to have sex with him, or not cooking properly, and 0 otherwise. All regressions control for religion, birth year, and state fixed effects.

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Appendix Mechanisms

Table Effects of the College Expansion on Women's Ownership

	House	Land	Phone	Bank Account
RD Estimate	0.081	0.081	0.032	0.018
	(0.018)	(0.016)	(0.015)	(0.017)
	[0.000]	[0.000]	[0.030]	[0.324]
Mean Dep Var	0.392	0.304	0.488	0.527
Bandwidth	1.371	1.565	2.974	2.703
Effective observations	14863	17093	33586	30241

Notes: Data is from NFHS-4. Each column reports results from a separate regression. *House*, *Land*, *Phone*, and *Bank Account* are indicators equal to 1 if the woman owns the corresponding asset (alone or jointly), and 0 otherwise. All regressions control for religion, birth year, and state fixed effects.

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Appendix Mechanisms

Table Effects of the College Expansion on Women's Participation in Household Decision

	Own Health Care	Major Purchases	Visits to Family	Husband's Earnings
RD Estimate	0.032	0.058	0.027	0.053
	(0.017)	(0.016)	(0.017)	(0.017)
	[0.015]	[0.000]	[0.030]	[0.000]
Mean Dep Var	0.755	0.738	0.750	0.712
Bandwidth	2.839	3.119	2.904	2.989
Effective observations	24310	26767	25100	25175

Notes: Data is from NFHS-4. Each column reports results from a separate regression. *Goes Out without Telling*, *Own Health Care*, *Major Purchases*, *Visits to Family*, and *Husband's Earnings* are indicators equal to 1 if the woman participates (alone or jointly) in decisions regarding her own health care, major household purchases, visits to her family or relatives, and the use of her husband's earnings, respectively, and 0 otherwise. All regressions control for religion, birth year, and state fixed effects.

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Appendix Mechanisms

Table Effects of the College Expansion on Women's Freedom of Mobility

	Allowed to go to market	Allowed to go to health facility	Allowed to go outside village
RD Estimate	0.021	0.025	0.013
	(0.011)	(0.009)	(0.008)
	[0.017]	[0.002]	[0.039]
Mean Dep Var	0.909	0.935	0.911
Bandwidth	1.833	1.931	3.494
Effective observations	19853	21191	40197

Notes: Data is from NFHS-4. Each column reports results from a separate regression. *Allowed to Go to Market*, *Allowed to Go to a Health Facility*, and *Allowed to Go Outside the Village* are indicators equal to 1 if the woman is allowed to go to the corresponding location alone, and 0 otherwise. All regressions control for religion, birth year, and state fixed effects.

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